

Tafsir Chowdhury

Full Stack AI Engineer

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SUMMARY

Backend-first Full Stack AI Engineer with system design depth, AI pipeline ownership, and DevOps working knowledge. 3+ years building production-grade web applications with TypeScript, Node.js, Next.js, and cloud infrastructure (AWS EC2/ECS, Railway). Proven track record architecting multi-LLM pipelines (OpenAI, Claude, Gemini), distributed worker systems, and scalable REST APIs for global clients. BSc in Computer Science with a 3.67 GPA and multiple IBM and Meta certifications.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, HTML5, CSS3, SQL

Frontend: React.js, Next.js, Tanstack Query, Zustand, Redux, Tailwind CSS, ShadCN UI, Ant Design

Backend: Node.js, Express.js, Drizzle, Prisma ORM, Mongoose, Redis, BullMQ, Better Auth, Auth.js, Clerk

AI & ML: LangChain, LangGraph, Vercel AI SDK, OpenAI API, Claude API, Gemini API, LangSmith

Databases: PostgreSQL, MongoDB, MySQL, Supabase, Neon.tech

DevOps & Cloud: AWS (EC2, ECS), Docker, GitHub Actions, DigitalOcean, Vercel, CI/CD Pipelines

Tools: Git, GitHub, Postman, Linux

PROFESSIONAL EXPERIENCE

Standard Insights

Oct '24 – Present

Full Stack AI Engineer

Remote | New York, USA

- Architected and shipped a multi-phase, multi-LLM AI report generation pipeline powering market trend analysis, consumer persona synthesis, and strategic insights – built as 32 durable Inngest functions with automatic retries, concurrency control, and failure callbacks, replacing a brittle single-request model and eliminating job loss on server restarts.
- Designed and owned a distributed background worker microservice processing AI workloads across 10+ concurrent queue types (quantitative data analysis, interview translation, persona generation, B2B scraping, Google Sheets sync); implemented batching and per-survey concurrency locks to prevent race conditions at scale.
- Built an AI-powered Survey Builder Engine with 7-intent discriminated-union Zod schema validation against Gemini structured output, enabling natural-language survey authoring and real-time LLM-to-state mutation with zero schema-drift between the API and database layer.
- Engineered a 6-step autonomous AI agent pipeline that ingests user-uploaded Excel/CSV files from AWS S3, validates format, parses multi-sheet structures, runs LLM-based schema analysis via Gemini, extracts survey questions and respondents, and re-uploads a normalized output file - all as idempotent, memoized steps with 3-retry fault tolerance.
- Built and optimized RESTful APIs across 50+ Express modules using MongoDB (50+ Mongoose models) and Redis caching, improving session state management and data-layer response consistency; implemented JWT-based auth, role-based access control, and Google OAuth2 token refresh lifecycle for third-party integrations (Google Sheets, Slack).
- Containerized 3 production services (client, server, admin) with Docker and managed deployments on AWS ECS, Railway, integrating CI/CD pipelines via GitHub Actions; deployed the worker service as an AWS Lambda function for cost-optimized serverless execution at scale.
- Delivered consumer intelligence dashboards and survey creation UIs using Next.js 15 App Router, TypeScript, TanStack Query, Zustand, and ShadCN/Radix UI, contributing to a Turborepo monorepo spanning 4 apps and 19 shared packages, reducing cross-app code duplication by an estimated ~40%.

HJ Animation Shop LTD

Full-Stack Developer

Jul '23 – Sep '24

Remote | Loughton, United Kingdom

- Built autonomous AI customer support platform (Aximus) using Next.js and Vercel AI SDK – engineered a ToolLoopAgent with dynamic tool-calling and a keyword-scoring RAG engine (no embedding API; 98% accuracy), with HMAC-SHA256 Meta webhook ingestion and atomic DB idempotency locks.
- Architected multi-tenant SaaS platforms in Next.js 16 and Drizzle ORM, including a floating-point-free commission math engine, custom binary JPEG/PNG metadata parser for AI portrait editing, and randomized exam engine with practical video submission grading.
- Integrated Better Auth with RBAC, 6-digit OTP 2FA, and session revocation across products; wired SumUp and UddoktaPay v2 payment flows, and serverless jsPDF routes with font-scaling for dynamic certificate generation.
- Delivered headless CMS (Parocia) with React Server Components (zero client JS on public pages), dynamic sitemap/robots.txt/JSON-LD schema compilation, UUID-hashed file storage with path-traversal protection.

PROJECTS

Visual Source | *Real-time soccer match operations and social graphic automation*

Live | 2026

- Architected a multi-tenant soccer club dashboard using Next.js 16, Turborepo, and Better-Auth, securing isolated workspaces for multiple sports organizations.
- Engineered a headless canvas rendering pipeline with Fabric.js, node-canvas, and Sharp to compile dynamic matchday graphics stored in Cloudflare R2.
- Integrated Facebook, Instagram, and Twitter Graph APIs, executing a polling and media chunking pipeline for real-time automated matchday posting.
- Designed a resilient Sportmonks API data synchronizer using PostgreSQL advisory locks to prevent database write collisions during concurrent syncing.
- Developed a live event logging UI with game-state tracking, automatic score override mechanisms, and a cron-scheduled graphic publishing queue.
- Deployed a containerized production stack using Docker Compose, orchestrating PostgreSQL, Next.js, and Caddy to automate SSL/HTTPS certificates and reverse proxying.

Everything Green | *Digital Sustainability & SEO Platform*

Live | 2025

- Designed a Retrieval-Augmented Generation (RAG) chatbot using MongoDB and OpenAI embeddings, featuring dynamic HTML scraping, semantic chunking, and context-aware query optimization to deliver digital sustainability advice.
- Developed an AI SEO recommendation engine that matches Google Ads suggestions to scraped website pages via cosine similarity, classifying conversion performance using a custom-trained CatBoost model.
- Implemented a secure subscription billing system using the Factory Pattern to integrate Stripe and PayPal, supporting customer portals, dynamic coupons, and webhook validation.
- Implemented a dynamic website badge generation system using Puppeteer-based headless rendering, dynamically generating customized carbon certification badges from HTML templates and real-time site scores.
- Built a responsive Next.js 15 dashboard with interactive D3/Sankey data visualizations and developed a Manifest V3 Chrome Extension to inject performance auditing metrics directly into Google SERPs.

Career Dock | *A Modern Job tracking application*

Live | 2024

- Built an end-to-end Next.js and Express.js web application utilizing Redux Toolkit (RTK Query) for cached state management, reducing client-side API latency by 30%.
- Integrated the Gemini API to construct prompt-chaining workflows that generate tailored resumes, LinkedIn profiles, and cover letters based on target job descriptions.
- Architected secure authentication and subscription monetization pipelines using Clerk for multi-tenant identity verification and Lemon Squeezy for premium plan billing.
- Designed interactive analytical dashboards using real-time charting components powered by dynamic Mongoose aggregations to track job application success rates.
- Deployed the distributed architecture to Vercel and cloud servers with a fully automated CI/CD pipeline, ensuring zero-downtime releases and instant preview deployments.

EDUCATION

University of the People

Bachelor of Science in Computer Science

2021 – 2024

DEAC | California, USA

- Graduated with CGPA 3.67
- Enlisted for Dean's Award List

CERTIFICATIONS

IBM: Full Stack Software Developer Specialization

Meta: Front-End Developer Specialization | Back-End Developer Specialization

Google: IT Support Specialization

Amazon Web Services: AWS Fundamentals Specialization

LANGUAGES

English: (Comfortable - professional and technical communication)

Bengali: (Native)

Hindi: (Communicative)